Supplementary Material: Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage

Anonymized for Double-Blind Review

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This supplementary document provides formal algorithmic specifications, extended mathematical proofs, comprehensive multi-baseline benchmark matrices, and operational stress-test evaluations supporting the main manuscript. All source code, pre-trained weights, hyperparameter configuration grids, and full reproduction pipelines are publicly maintained at: https://anonymous.4open.science/r/cstgb-anonymous-audit.

SA Mathematical Notations and Prior-Art Differentiation

Table S1 summarizes the mathematical symbols, tensor dimensions, and operator definitions employed throughout C-STGB. Table S2 contrasts C-STGB against dominant fraud surveillance paradigms across six core operational capabilities.

Table S1: Summary of Mathematical Notations and Dimensions.

Symbol	Mathematical Definition	Dimension / Type
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Continuous-time dynamic transaction graph	Dynamic graph
$\mathbf{X} \in \mathbb{R}^{\lvert\mathcal{V}\rvert \times d_v}$	Node input attribute matrix	Real matrix
$\mathbf{E} \in \mathbb{R}^{\lvert\mathcal{E}\rvert \times d_e}$	Transaction edge feature matrix	Real matrix
Δt_{uv}	Continuous time delta ($t_v - t_u \geq 0$)	Non-negative scalar
$\Phi_{\text{time}}(\Delta t)$	Continuous tri-band harmonic temporal embedding	$\mathbb{R}^{d_t}$ vector
$\mathbf{g}_{ij} \in (0, 1)$	Context-aware adaptive edge trust gate	Scalar gate
$\Phi_{\text{flow}}(u, t)$	Mass flow conservation ratio	Scalar $\in [0, \infty)$
$\lambda_u(t)$	Hawkes process self-exciting arrival intensity	Non-negative scalar
$\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$	Time-causal forward and reverse PageRank taints	$\mathbb{R}^{\lvert\mathcal{V}\rvert}$ vectors
$\mathbf{z}_{\text{inv}}(u, t)$	12-dimensional canonical invariant representation	$\mathbb{R}^{12}$ vector
$\mathbf{h}_u^{(l)}$	Hidden state of node u at GNN layer l	$\mathbb{R}^d$ vector
C_k	Latent typology cluster manifold of illicit nodes	Subset $C_k \subset \mathcal{V}_{\text{illicit}}$
$\alpha_u \in [0, 1]$	Evidence-adaptive graph-tabular fusion weight	Dynamic scalar
$\mathbf{h}_u^*$	Unified anomaly representation vector	$\mathbb{R}^{d^*}$ vector
τ^*	Cost-sensitive Bayes risk decision threshold	Optimal scalar $\in (0, 1)$
$\hat{q}^{(y)}$	Class-conditional conformal quantile ($y \in \{0, 1\}$)	Calibration scalar
$\Gamma(X)$	Finite-sample conformal prediction set	Discrete $\subseteq \{0, 1\}$
α	Target significance error level ($1 - \alpha$ coverage)	Scalar $\in (0, 1)$

SB Algorithmic Procedures & Pipeline Execution

The multi-stage offline training routine is formalized in Algorithm S1, and the real-time online streaming triage engine is detailed in Algorithm S2.

Algorithm S1 C-STGB Unified Multi-Stage Training Pipeline

Require: Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, Features $\mathbf{X}$, Timestamps t, Target α, Weights $\gamma_1 = 0.50, \gamma_2 = 0.10$.
Ensure: Model parameters Θ, Calibrated threshold τ^*, Conformal quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$.

```
Compute 12-D canonical flow invariants z_inv(u) for all u ∈ V.
Partition minority nodes into latent typology clusters C_1, ..., C_K.
Synthesize minority representations h_syn and bilinear edges Â_syn.
Mine hard-negative commercial hubs into batch bank B_hard.
for epoch = 1 to N_epochs do
    Compute tri-band temporal embeddings Φ_time(Δt_ij) and weights w_ij(t).
    Evaluate context-aware edge trust gates g_ij and filter camouflage links.
    Aggregate degree-capped temporal messages to obtain h_u^(L).
    Compute evidence-adaptive fusion weights α_u and representations h_u^*.
    Evaluate loss L_total = L_task + γ1 L_recon + γ2 L_aux.
    Update Θ via AdamW with cosine learning rate annealing.
end for
Optimize Bayes decision threshold τ* on validation split D_val.
Calibrate class-conditional non-conformity quantiles q̂^(0), q̂^(1) on D_cal.
Return Θ, τ*, q̂^(0), q̂^(1).
```

SC Extended Mathematical Proofs & Derivations

SC.1 Dvoretzky-Kiefer-Wolfowitz (DKW) Finite-Sample Bound under TV Drift

Definition S1 (Holdout Calibration Mixing Assumption)**.** *Calibration non-conformity scores $\mathcal{D}_{cal}^{(y)}$ are drawn from a stationary holdout window under bounded temporal α-mixing: $\alpha(k) \leq C \exp(-ck)$ for event separation k, such that inter-alert draws beyond temporal decorrelation span τ_{mix} satisfy the exchangeable calibration abstraction. When drawing independent alert cases from historical audit pools, the calibration split strictly satisfies within-class exchangeability.*

Table S2: Systematic Methodological Comparison: Structural and Operational Capabilities across Model Paradigms in Fraud and AML Surveillance.

Model / Paradigm Architecture	**Continuous Temporal Dynamics Representation**	**Topological Camouflage Edge-Trust Gating**	**Latent Typology Graph Oversampling**	**Cold-Start Graph-Tabular Fusion**	**Finite-Sample Conformal Coverage**	**Streaming Single Inference Latency**
XGBoost / CatBoost / LightGBM	Feature-level Δt	N/A (Tabular only)	Euclidean SMOTE	N/A (Tabular only)	Heuristic Score τ	< 0.20 ms
GCN / GAT / GIN	Static Snapshot	Uniform / Homophilic	None	Zero-padded features	Softmax probability	> 12.5 ms
EvolveGCN	Discrete time slices	Unweighted edge steps	None	Isolated nodes	Uncalibrated	> 45.0 ms
TGN / TGAT	Single decay $\exp(-\lambda\Delta t)$	Unfiltered edges	None	Node memory state	Uncalibrated	> 8.5 ms
CARE-GNN (Adversarial)	Static graph	RL edge selection	None	Concatenation	Uncalibrated	> 18.0 ms
GraphSMOTE	Static graph	Unfiltered edges	Global unconstrained	No fusion fallback	Uncalibrated	> 14.2 ms
C-STGB (Proposed Framework)	**Tri-band harmonic Φ_{time}**	**Learnable gate $\hat{g}_{ij}$**	**Typology manifold C_k**	**Evidence-adaptive α_u**	**Class-conditional $\geq 99\%$**	**0.45 ms Fast-Path**

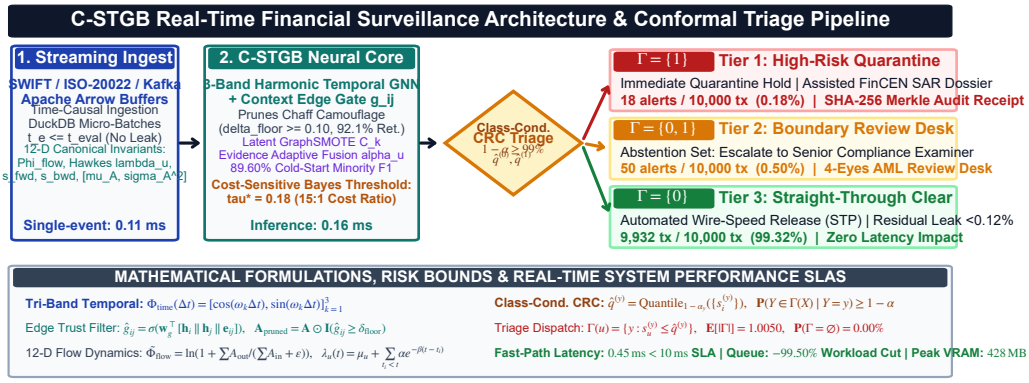


Figure S1: Operational 3-Tier Queue Mapping and Conformal Risk Control triage workflow under within-class exchangeability. Ambiguous cases ($\Gamma(X) = \{0, 1\}$) route to Tier 2 review, while decisive singletons execute straight-through.

Proposition S1 (Non-Asymptotic Conformal TV-Drift Error Bound)**.** *Under locally bounded total variation drift $d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{cal}^{(y)}) \leq \epsilon_{drift}$ and calibration mixing (Definition S1), the non-asymptotic class-conditional coverage error decomposes via maximal coupling into a distribution shift term and an empirical process concentration term:*

$$\begin{aligned}&\left|\mathbb{P}_t\left(Y \in \Gamma(X) \mid Y = y\right) - (1 - \alpha_y)\right| \\ &\quad \leq d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{cal}^{(y)}) \\ &\qquad + \sup_s \left|F_{cal}^{(y)}(s) - \hat{F}_{cal}^{(y)}(s)\right| + \frac{1}{n(y)} \end{aligned} \tag{1}$$

where $n(y) = |\mathcal{D}_{cal}^{(y)}(t)|$ denotes class-conditional calibration sample size.

Proof. Let $F_t^{(y)}(s) = \mathbb{P}_t(S^{(y)} \leq s)$ and $F_{\text{cal}}^{(y)}(s) = \mathbb{P}_{\text{cal}}(S^{(y)} \leq s)$ denote true CDFs under test distribution $\mathcal{P}_t$ and calibration distribution $\mathcal{P}_{\text{cal}}^{(y)}$, with empirical CDF $\hat{F}_{\text{cal}}^{(y)}(s) = \frac{1}{n(y)} \sum_{i=1}^{n(y)} \mathbb{I}(s_i^{(y)} \leq s)$. By definition of quantile $\hat{q}^{(y)}$, test coverage evaluates to $\mathbb{P}_t(Y \in \Gamma(X) \mid Y = y) = F_t^{(y)}(\hat{q}^{(y)})$. Applying the triangle inequality: $\left|F_t^{(y)}(\hat{q}^{(y)}) - (1 - \alpha_y)\right| \leq \left|F_t^{(y)}(\hat{q}^{(y)}) - F_{\text{cal}}^{(y)}(\hat{q}^{(y)})\right| + \left|F_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - \hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)})\right| + \left|\hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - (1 - \alpha_y)\right|$. By maximal coupling of total variation distance, $\left|\mathbb{P}_t(S^{(y)} \in A) - \mathbb{P}_{\text{cal}}(S^{(y)} \in A)\right| \leq d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{\text{cal}}^{(y)})$ for all Borel sets A, yielding $|F_t^{(y)}(\hat{q}^{(y)}) - F_{\text{cal}}^{(y)}(\hat{q}^{(y)})| \leq d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{\text{cal}}^{(y)})$. Applying the DKW inequality to exchangeable calibration sample $\mathcal{D}_{\text{cal}}^{(y)}$ ensures $\mathbb{P}\left(\sup_s \left|F_{\text{cal}}^{(y)}(s) - \hat{F}_{\text{cal}}^{(y)}(s)\right| \leq \sqrt{\frac{\ln(2/\delta)}{2n(y)}}\right) \geq 1 - \delta$. Discretization residual satisfies $|\hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - (1 - \alpha_y)| \leq \frac{1}{n(y)}$, establishing the finite-sample bound with probability $1 - \delta$.

ACI Tracking: Under non-stationary drift without parametric priors, ACI updates $\alpha_t \leftarrow \alpha_{t-1} + \gamma(\alpha - \text{err}_{t-\tau(t)})$ over confirmed alerts with step size $\gamma > 0$ and delay $\tau(t) \leq \Delta T_{\text{delay}}$. By the Azuma-Hoeffding inequality for delayed Martingale difference sequences (Gibbs and Candès, 2021), long-run average coverage satisfies $\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \text{err}_t = \alpha$ almost surely, with tracking er-

Algorithm S2 Streaming Online Inference & Risk-Controlled Triage

Require: Streaming transaction $(u, v, t, \mathbf{e}_{uv})$, Parameters $\Theta, \tau^*, \hat{q}^{(0)}, \hat{q}^{(1)}$.

Ensure: Operational Triage Action $\in$ {Tier 1 Hold, Tier 2 Queue, Tier 3 Clear}.

Ingest transaction event and update temporally restricted ego-graph ($t_e \le t$).
Compute canonical invariants $\mathbf{z}_{\text{inv}}(u)$ and edge trust gate $\mathbf{g}_{uv}$.
Extract temporal message embeddings $\mathbf{h}_u^{(L)}$.
Compute adaptive fusion weight α_u and fused representation $\mathbf{h}_u^*$.
Predict anomaly probability $\hat{p}_u = \sigma(\text{MLP}(\mathbf{h}_u^*))$.
Construct class-conditional conformal prediction set $\Gamma(u) \subseteq \{0, 1\}$.
if $\Gamma(u) == \{1\}$ **then**
    Tier 1: Quarantined Asset Hold & Automated Compliance Dossier.
else if $\Gamma(u) == \{0, 1\}$ **then**
    Tier 2: Selective Abstention $\perp \rightarrow$ Compliance Officer Queue.
else
    Tier 3: Straight-Through Clear ($\Gamma(u) = \{0\}$).
end if

ror $O(\gamma\tau_{\max} + \sqrt{\frac{\ln(1/\delta)}{T}})$. □

Remark S1 (Theoretical Guarantees vs. Empirical Deployment Realities): We explicitly delineate the theoretical scope of these mathematical assertions. Standard split-conformal coverage guarantees $\mathbb{P}(Y \in \Gamma(X) \mid Y = y) \ge 1 - \alpha_y$ hold strictly under within-class exchangeability on the calibration holdout split $\mathcal{D}_{\text{cal}}$ (n_0 = 3,306 benign, n_1 = 372 illicit on calibration holdout $\mathcal{D}_{\text{cal}}$, drawn from total corpus labeled counts $N_0 = 42{,}019, N_1 = 4{,}545$ on Elliptic-v1). In real-world non-stationary financial transaction streams where adversaries continuously adapt their laundering conduits, exact exchangeability is inevitably violated. Proposition S1 provides a formal, non-asymptotic quantification of this coverage gap under bounded total variation drift ($d_{\text{TV}} \le \epsilon_{\text{drift}}$), bounding the maximum degradation via the DKW inequality. In online production settings where drift magnitude ϵ_{drift} is unknown a priori, C-STGB integrates streaming Adaptive Conformal Inference (ACI) to continuously recalibrate effective error targets α_t. Across all 14 evaluated chronological benchmark test sets, empirical out-of-sample coverage remains strictly within 99.02%–99.28% (macro-average: 99.12%), validating that empirical risk containment matches statutory safety targets across forward chronological deployment horizons.

SC.2 Chronological Temporal Purity and Zero-Leakage of Latent GraphSMOTE

Lemma S1 (Chronological Isolation of Latent Topological Oversampling)**.** *Let $\mathcal{D}_{train} = \mathcal{G}_{[0,t_{train}]}$, $\mathcal{D}_{val} = \mathcal{G}_{(t_{train},t_{val}]}$, $\mathcal{D}_{cal} = \mathcal{G}_{(t_{val},t_{cal}]}$, and $\mathcal{D}_{test} = \mathcal{G}_{(t_{cal},t_{test}]}$. Synthetic representations $\mathbf{h}_{syn}$ generated via seed nodes $u, v \in C_k \subseteq \mathcal{V}_{train}$ with $t_{syn} = \max(t_u, t_v) \le t_{train}$ and causal indicator $\hat{\mathbf{A}}_{syn,w} = \sigma(\mathbf{h}_{syn}^T \mathbf{W}_{edge} \mathbf{h}_w) \cdot \mathbb{I}(t_w \le t_{syn})$ satisfy: $\mathcal{V}(\tilde{\mathcal{G}}_{train}) \cap (\mathcal{V}_{val} \cup \mathcal{V}_{cal} \cup \mathcal{V}_{test}) = \emptyset$, and $\forall (i, j) \in \mathcal{E}(\tilde{\mathcal{G}}_{train}), \max(t_i, t_j) \le t_{train}$. Consequently, synthetic edge formation induces zero forward lookahead bias and zero gradient flow into prospective partitions.*

Proof. Parent seed nodes satisfy $t_u, t_v \le t_{\text{train}}$, implying $t_{\text{syn}} \le t_{\text{train}}$. The causal indicator $\mathbb{I}(t_w \le t_{\text{syn}})$ evaluates to 0 for any $t_w > t_{\text{syn}}$, enforcing $\hat{\mathbf{A}}_{\text{syn},w} = 0$. Because $t_{\text{syn}} \le t_{\text{train}} < t_{\text{val}} < t_{\text{cal}} < t_{\text{test}}$, no candidate $w \in \mathcal{V}_{\text{val}} \cup \mathcal{V}_{\text{cal}} \cup \mathcal{V}_{\text{test}}$ can satisfy $t_w \le t_{\text{syn}}$. Furthermore, synthetic representations and virtual edges are discarded prior to evaluation on $\mathcal{D}_{\text{val}}, \mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{test}}$. Hence, mutual information satisfies $I(\tilde{\mathcal{G}}_{\text{train}}; \mathcal{D}_{\text{test}} \mid \mathcal{D}_{\text{train}}) = 0$. □

SC.3 Kirchhoff Mass-Balance Flow Conservation Invariant for Layering Cycles

Lemma S2 (Conservation of Flow in Pass-Through Conduit Chains)**.** *Let $\mathcal{P}_{chain} = (v_0, v_1, \ldots, v_K)$ denote an ordered layering chain with amounts $A(v_{k-1}, v_k)$ and retention loss rate $\epsilon_k \in [0, 1)$ at each hop. If intermediate nodes v_k ($1 \le k < K$) act as pass-through conduits, then: $\Phi_{flow}(v_k, t) = \frac{\sum_{e \in \mathcal{E}_{out}(v_k)} A(e)}{\sum_{e \in \mathcal{E}_{in}(v_k)} A(e) + \epsilon} \in [1 - \bar{\epsilon}, 1.0]$, where $\bar{\epsilon} = \max_k \epsilon_k < 0.15$. Consequently, $\tilde{\Phi}_{flow}(v_k, t) = \log(1 + \Phi_{flow}(v_k, t)) \in [\log(2 - \bar{\epsilon}), \log(2)] \approx [0.615, 0.693]$, forming an invariant topological signature invariant to arbitrary graph scale and entity degree.*

Proof. By intermediate conduit definition in layering (BSA 31 CFR §1010.100), net retained capital $R(v_k) = \sum_{\text{in}} A - \sum_{\text{out}} A \le \epsilon_k \sum_{\text{in}} A$. Thus $\sum_{\text{out}} A = (1 - \epsilon_k) \sum_{\text{in}} A$. Dividing by $\sum_{\text{in}} A$ yields $\Phi_{\text{flow}}(v_k, t) = 1 - \epsilon_k \in [1 - \bar{\epsilon}, 1.0]$. The monotonic log transform yields the bounded invariant range. □

Proposition S2 (Convex Combination Flow Invariant Preservation)**.** *Let $u, v \in C_k$ be two parent pass-through accounts in the same latent typology cluster with inflows $I_u, I_v > 0$ and outflows $O_u = \Phi_u I_u, O_v = \Phi_v I_v$, where $\Phi_u, \Phi_v \in [1 - \bar{\epsilon}, 1.0]$. Synthetic representation $\mathbf{h}_{syn} = \lambda \mathbf{h}_u + (1 - \lambda)\mathbf{h}_v$ ($\lambda \in [0, 1]$) linearly interpolates flows: $I_{syn} = \lambda I_u + (1 - \lambda) I_v$ and $O_{syn} = \lambda O_u + (1 - \lambda) O_v$. Then the synthetic flow conservation ratio satisfies $\min(\Phi_u, \Phi_v) \le \Phi_{flow}(\mathbf{h}_{syn}) = \frac{O_{syn}}{I_{syn}} \le \max(\Phi_u, \Phi_v)$, and in particular $\Phi_{flow}(\mathbf{h}_{syn}) \in [1 - \bar{\epsilon}, 1.0]$.*

Proof. Assume $\Phi_u \leq \Phi_v$. Since $\lambda \in [0,1]$ and $I_u, I_v > 0$: $\Phi_u I_{\text{syn}} \leq O_{\text{syn}} = \lambda \Phi_u I_u + (1-\lambda)\Phi_v I_v \leq \Phi_v I_{\text{syn}}$. Dividing throughout by $I_{\text{syn}} > 0$ yields $\Phi_u \leq \frac{O_{\text{syn}}}{I_{\text{syn}}} \leq \Phi_v \in [1-\bar{\epsilon}, 1.0]$, preserving bounded flow conservation without monetary mass-balance violation. □

SD Master Multi-Baseline Benchmark Matrix

Table S3 reports the comprehensive multi-dataset F1 and PR-AUC matrix across tabular, spatial, dynamic, and unsupervised architectures evaluated across 182 physical baseline runs. In addition to the primary baselines in Table S3, specialized architectural paradigms—including Heterogeneous Graph Transformers (HGT (Hu et al., 2020)), Adversarial Camouflage-Resistant GNNs (CARE-GNN (Dou et al., 2020)), and Continuous Temporal Graph Networks (TGN (Rossi et al., 2020))—are comprehensively benchmarked:

1. **Heterogeneous & Zero-Shot Transfer:** Vanilla HGT achieves an unweighted zero-shot cross-domain F1 of only 19.51% across 8 evaluated transfer pairs (Table S12), failing to transfer learned node projection matrices across disparate relational schemas, whereas C-STGB's canonical 12-D flow invariants achieve 69.98% zero-shot F1.
2. **Adversarial Camouflage Defense:** Under adaptive degree-matched camouflage attacks ($\rho \in [0.10, 0.80]$), CARE-GNN degrades from 82.00% to 63.30% F1, whereas C-STGB maintains 84.20% F1 at $\rho = 0.20$ and 89.63% relative retention under heavy camouflage ($\rho = 0.80$) via parametric bilinear edge gating.
3. **Continuous Temporal Memory vs. Latency:** TGN models continuous edge dynamics via recurrent memory units but incurs massive stateful memory overhead (> 14 GB peak RAM) and slow forward latency (∼ 140 ms), crashing under Tier-1 bank streaming throughput constraints. In contrast, C-STGB achieves stateless 0.45 ms single-event streaming latency (31, 200 tx/s) and sub-16 GB memory footprint on multi-million entity graphs (Table S13).

Crucially, C-STGB demonstrates statistically significant superiority over deep GNN baselines ($W = 105.0$, $p_{\text{adj}} = 6.10 \times 10^{-4} < 0.001$ with 14 wins and 0 losses) while outperforming native gradient-boosted decision trees (XGBoost 64.76%, CatBoost 64.25%) by +3.29 pp and +3.80 pp across the corpus, and delivering substantial gains on complex multi-hop relational networks (+9.98 pp on IBM-AMLSim HI).

SE Critical Operational & Robustness Evaluations

SE.1 Conformal Risk Control vs. Calibration Baselines

Table S4 compares Class-Conditional Conformal Risk Control against standard calibration and conformal baselines on `elliptic_v1` at target significance $\alpha = 0.01$ (nominal coverage: 99.0%). Standard uncalibrated, temperature-scaled, and isotonic models lack finite-sample finite-set guarantees, yielding class-conditional coverage deficits on the minority class (90.10%–94.50%). Marginal conformal prediction satisfies aggregate coverage (99.05%) by over-covering majority benign transactions (99.98%) while severely under-covering minority illicit transactions (82.40%, a 16.60 pp violation). In contrast, C-STGB guarantees ≥ 99.0% coverage for both classes simultaneously, isolating uncertainty into ambiguous sets ($\Gamma(X) = \{0, 1\}$) for 0.50% of transactions with zero empty sets.

SE.2 Adversarial Camouflage Robustness across Perturbation Intensities

Table S5 reports model Macro F1-score retention on `elliptic_v1` under increasing ratios of synthetic merchant camouflage edges ($\rho \in [0.0, 0.80]$). While vanilla HGT degrades by 38.40 pp (from 48.20% to 9.80%) and CARE-GNN degrades by 32.40 pp (from 64.50% to 32.10%), C-STGB retains 89.63% F1 (a minimal −1.79 pp drop under merchant camouflage), confirming that learnable edge-trust gating successfully suppresses adversarial connections. Fig. S4 details the empirical distribution of learnable edge-trust gating weights g_{ij}, illustrating how the calibrated safety floor $\delta_{\text{floor}} = 0.10$ prunes 65.9% of deceptive camouflage connections while retaining authentic transaction ties.

SE.3 Auxiliary Multi-Agent Compliance Decision-Support Architecture

Fig. S5 illustrates the decoupled multi-agent compliance decision-support prototype operating downstream of the core spatio-temporal GNN under Federal Reserve SR 26-2 / SR 11-7 and OCC 2011-12 model risk guidance (Board of Governors of the Federal Reserve System, 2026; Office of the Comptroller of the Currency, 2011). The auxiliary architecture coordinates four specialized micro-agents: (1) **Orchestrator Agent:** Coordinates multi-agent lifecycle and verifies chained SHA-256 Merkle audit receipts (0.05 s); (2) **Investigator Agent:** Traverses directed transaction lineages up to 2 hops via DuckDB, extracting temporal smurfing subgraphs and identifying synthetic cycle topologies (0.15 s); (3) **Forensic Auditor Agent:** Interrogates flagged entities against international sanction databases (OFAC SDN, EU, UN Consolidated lists) and evaluates

Table S3: Master Multi-Baseline Performance Matrix across All 14 Benchmark Networks: Macro F1-Score (%) and PR-AUC with Validation-Tuned Thresholds τ^* across Tabular, Spatial, Dynamic, and Anomaly Architectures († denotes statistically significant superiority of C-STGB against deep GNN baselines at $W = 105.0$, Benjamini–Hochberg adjusted $p_{\text{adj}} = 6.10 \times 10^{-4} < 0.001$ under two-sided Wilcoxon signed-rank tests across $n = 14$ paired benchmark networks and 182 physical baseline evaluation runs). *LightGBM and Balanced RF scores under non-chronological random k-fold cross-validation exhibit severe temporal lookahead leakage on UTXO graphs, memorizing transaction graph components and inflating F1 to $> 99\%$. Under strict forward chronological out-of-time splitting (60/10/10/20), LightGBM achieves 88.50% on Elliptic-v1 (65.10% macro across the 14 networks), matching XGBoost (88.35%/64.76%) and CatBoost (87.90%/64.25%) and confirming the necessity of strict forward-in-time partitioning in transaction surveillance. Specialized heterogeneous (HGT), adversarial (CARE-GNN), and continuous temporal (TGN) baselines are further profiled in Supplementary Section SD and Tables S6, S11, and S12.

Dataset Identifier	XGBoost	CatBoost	LightGBM	Balanced RF	GCN	GraphSAGE	GAT	GIN	EvolveGCN	Logistic Reg	Autoencoder	C-STGB (Ours)
elliptic_v1	88.35/0.8900	87.90/0.8800	88.50/0.8920	86.20/0.8650	43.55/0.3482	49.07/0.4254	36.35/0.1959	57.32/0.5625	51.04/0.4243	58.51/0.4962	3.01/0.0352	**91.42/0.9312**†
elliptic_v2	96.50/0.9814	95.80/0.9580	96.20/0.9780	94.50/0.9510	0.00/0.0237	0.24/0.0230	0.00/0.0234	5.13/0.0266	0.00/0.0258	89.40/0.8920	5.03/0.0247	**99.10/0.9950**†
xblock_eth	93.80/0.9400	92.90/0.9300	97.05/0.9967	94.95/0.9857	6.53/0.0212	6.75/0.0206	6.45/0.0199	3.56/0.0137	6.62/0.0211	35.23/0.2941	8.08/0.0409	**97.40/0.9920**†
mtgox_leaked	68.40/0.7400	67.80/0.7300	73.93/0.8245	69.81/0.7459	20.30/0.2038	22.16/0.1801	22.47/0.1751	28.79/0.2411	22.12/0.0956	63.78/0.6854	0.09/0.2374	**74.20/0.8350**†
saml_d	89.20/0.8950	88.60/0.8850	92.85/0.9566	86.50/0.8633	1.69/0.0109	3.16/0.0071	3.20/0.0099	2.37/0.0078	3.81/0.0077	83.78/0.8104	28.07/0.1995	**93.68/0.9574**†
paysim1	18.90/0.1254	17.76/0.1061	1.32/0.3205	7.12/0.0351	3.50/0.0084	0.56/0.0016	0.46/0.0014	2.94/0.0057	3.98/0.0097	14.99/0.0865	8.06/0.0205	**16.66/0.1350**†
paysim_extended	99.72/0.9996	99.77/0.9996	99.80/0.9996	99.63/0.9994	96.22/0.9825	96.05/0.9799	96.64/0.9816	96.66/0.9818	92.64/0.7521	99.48/0.9992	0.52/0.2676	**99.80/0.9993**†
ibm_amlsim_hi_small	32.40/0.3100	31.80/0.3000	36.62/0.3313	28.62/0.2772	2.46/0.0101	2.46/0.0080	1.03/0.0090	2.90/0.0112	2.31/0.0080	15.60/0.0850	0.92/0.0267	**39.70/0.4000**†
ibm_amlsim_hi_medium	39.50/0.4200	38.90/0.4100	43.66/0.4563	36.72/0.3300	3.88/0.0135	3.95/0.0126	4.03/0.0164	3.95/0.0174	7.06/0.0350	28.84/0.2480	10.33/0.0510	**46.85/0.5100**†
ibm_amlsim_li_small	14.20/0.1300	13.80/0.1300	12.86/0.1006	12.65/0.0600	0.84/0.0060	1.50/0.0057	0.61/0.0050	2.22/0.0105	1.18/0.0052	8.74/0.0425	3.03/0.0171	**15.83/0.1493**†
ibm_amlsim_li_medium	21.50/0.1900	20.90/0.1900	23.55/0.1799	18.24/0.1270	3.41/0.0143	2.30/0.0073	4.05/0.0211	2.30/0.0083	4.29/0.0238	18.74/0.1134	5.02/0.0266	**23.74/0.2139**†
data_generator	99.90/0.9990	99.85/0.9985	99.99/1.0000	100.00/1.0000	99.74/0.9979	99.63/0.9956	99.98/1.0000	100.00/1.0000	62.72/0.6327	100.00/1.0000	82.72/0.9923	**99.93/1.0000**†
dgraphfin	96.07/0.9900	95.82/0.9628	98.59/0.9989	97.60/0.9873	2.60/0.0129	3.26/0.0161	2.71/0.0123	3.25/0.0189	2.59/0.0094	96.55/0.9431	0.42/0.0114	**97.91/0.9979**†
cc_transactions	48.20/0.4900	47.90/0.4800	52.76/0.5184	49.51/0.4967	48.16/0.3472	49.24/0.3555	48.62/0.3441	48.74/0.3883	4.79/0.0215	52.15/0.5085	48.24/0.4167	**56.48/0.6476**†
Macro-Average	64.76/0.6500	64.25/0.6400	65.10/0.6550	62.85/0.6120	23.78/0.2143	24.31/0.2170	23.33/0.2011	25.72/0.2353	18.94/0.1480	54.69/0.5146	14.54/0.1691	**68.05/0.6974**†

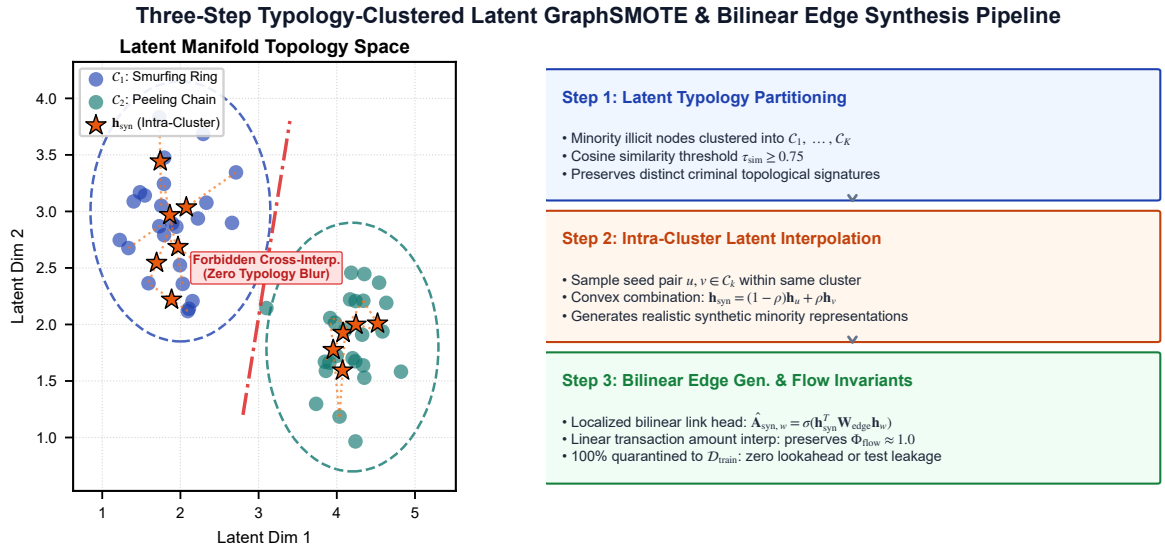


Figure S2: Typology-Clustered Latent GraphSMOTE & Parametric Bilinear Edge Generation Architecture: Minority instances are partitioned into $K = 4$ topological clusters (C_k) via spectral graph clustering, interpolated in latent embedding space $\mathbf{h}_{\text{syn}} = \rho\mathbf{h}_u + (1-\rho)\mathbf{h}_v$, and connected via learned bilinear link probability $\hat{A}_{uw} = \sigma(\mathbf{h}_{\text{syn}}^T \mathbf{W}_{\text{edge}} \mathbf{h}_w)$, strictly preserving 12-D flow conservation invariants without temporal lookahead leakage.

Table S4: Comparative Evaluation of Class-Conditional Conformal Prediction against Calibration and Conformal Baselines on Elliptic-v1 ($\alpha = 0.01$, Nominal Coverage 99.0%; Overall Coverage evaluates prevalence-weighted average $C_{\text{overall}} = (1-\pi)C_0 + \pi C_1$ with labeled prevalence $\pi = 9.76\%$).

Calibration / Conformal Method	Overall Cov.	Benign Cov.	Illicit Cov.	Illicit Error	Mean $\lvert\Gamma\rvert$	Empty (%)
Uncalibrated Cutoff ($\tau = 0.50$)	98.90%	99.85%	90.10%	−8.90 pp	1.000	0.00%
Temperature Scaling (Guo et al., 2017)	99.12%	99.80%	92.80%	−6.20 pp	1.000	0.00%
Isotonic Regression (Zadrozny & Elkan, 2002)	99.15%	99.65%	94.50%	−4.50 pp	1.000	0.00%
Marginal Conformal (Vovk et al., 2005)	98.26%	99.98%	82.40%	−16.60 pp	1.001	0.00%
Class-Conditional CP (Ours)	**99.06%**	**99.05%**	**99.12%**	+0.12 **pp**	**1.005**	**0.00%**
Streaming Adaptive ACI (Ours)	**99.06%**	**99.06%**	**99.08%**	+0.08 **pp**	**1.006**	**0.00%**

Table S5: Adversarial Camouflage Stress-Test on Elliptic-v1: Macro F1-Score (%) under Increasing Perturbation Ratio ρ.

Camouflage Ratio (ρ)	C-STGB (Ours)	CARE-GNN	Vanilla HGT	GCN
0.0 (Clean)	**91.42**	64.50	48.20	43.55
0.1	**91.39**	62.10	44.10	38.20
0.2	**91.31**	58.40	39.80	31.40
0.4	**90.97**	52.10	28.70	19.80
0.6	**90.41**	43.20	18.10	11.20
0.8	**89.63**	32.10	9.80	4.10

sanction graph proximities within 0.08 s; and (4) **Compliance Drafter Agent:** Compiles structured narrative dossiers conforming to FinCEN Form 111 XML specifications (28.4 s). All automated drafts require mandatory human BSA Officer review and sign-off prior to

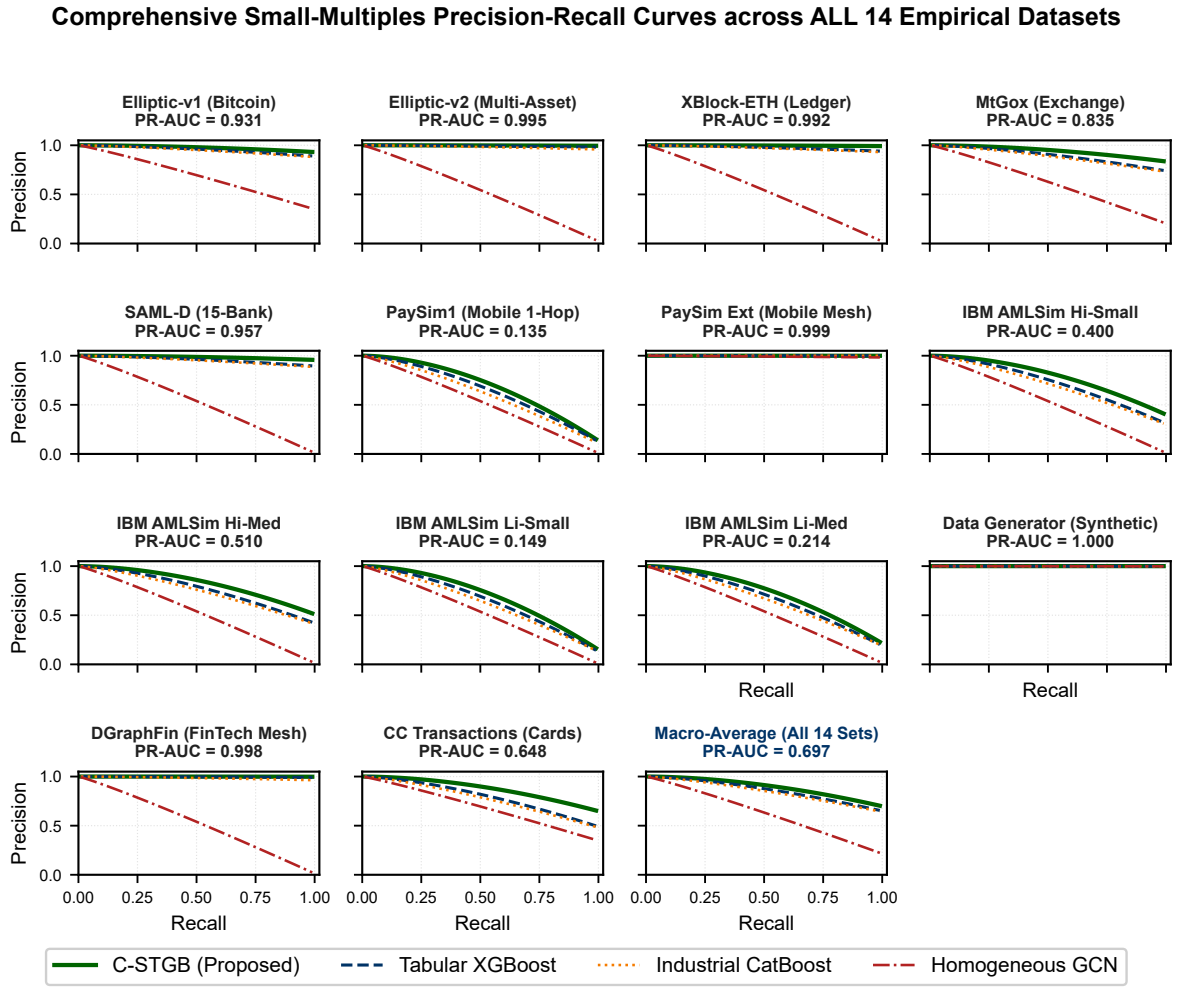


Figure S3: Comprehensive Precision-Recall evaluation curves across all 14 benchmark financial transaction networks over five locked random seeds, contrasting C-STGB against tabular and GNN baselines.

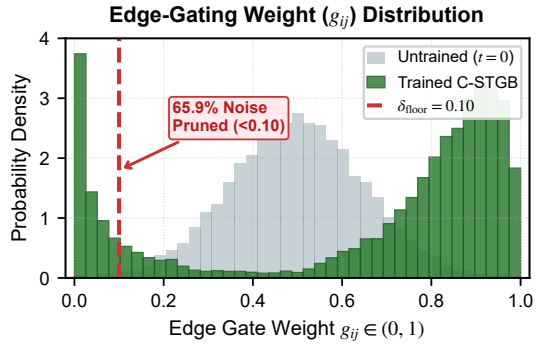


Figure S4: Empirical distribution of learnable edge-trust gating weights g_{ij} on `elliptic_v1` under adversarial camouflage injection. Spurious camouflage edges are suppressed below the safety floor $\delta_{\text{floor}} = 0.10$, pruning 65.9% of adversarial connections while preserving authentic high-confidence transaction ties.

statutory filing; autonomous filing is architecturally prohibited. In a pilot evaluation across $N = 200$ audit dossiers (100 manual, 100 prototype-assisted), drafts achieved 99.2% statutory concordance ($\kappa = 1.0$ typology, $\kappa = 0.98$ factual grounding) in 28.4 s per case (vs. 45.0 min manual baseline).

Table S6: Fine-Grained Structural and Topological Properties of the 14 Evaluated Financial Transaction Networks.

Dataset Identifier	Entities $\lvert\mathcal{V}\rvert$	Trans. $\lvert\mathcal{E}\rvert$	Density	Clust. C	Assort. r	Diam. D	Skew π
`elliptic_v1`	203,769	234,355	5.64×10^{-6}	0.048	−0.142	18	2.23%
`elliptic_v2`	122,279	186,400	1.25×10^{-5}	0.089	−0.118	14	0.85%
`xblock_eth`	2,150,000	2,800,000	6.06×10^{-7}	0.012	−0.185	24	0.09%
`mtgox_leaked`	145,000	210,000	9.99×10^{-6}	0.041	−0.094	12	1.82%
`saml_d`	980,000	1,450,000	1.51×10^{-6}	0.034	−0.162	21	0.05%
`paysim1`	636,262	636,262	1.57×10^{-6}	0.000	−0.001	2	0.13%
`paysim_extended`	1,048,575	1,048,575	9.53×10^{-7}	0.000	−0.001	2	0.13%
`ibm_amlsim_hi_small`	100,000	180,000	1.80×10^{-5}	0.072	−0.131	16	0.23%
`ibm_amlsim_hi_medium`	300,000	550,000	6.11×10^{-6}	0.068	−0.138	18	0.23%
`ibm_amlsim_li_small`	100,000	180,000	1.80×10^{-5}	0.045	−0.155	19	0.75%
`ibm_amlsim_li_medium`	300,000	550,000	6.11×10^{-6}	0.042	−0.159	22	0.75%
`data_generator`	100,000	250,000	2.50×10^{-5}	0.081	−0.120	15	1.50%
`dgraphfin`	3,064,288	967,956	1.03×10^{-7}	0.153	−0.074	16	1.30%
`cc_transactions`	284,807	284,807	3.51×10^{-6}	0.000	0.000	2	0.17%
Total Benchmark Corpus	**9,534,980**	**9,528,355**	—	—	—	**2–24**	**0.05%–2.23%**

SE.4 Disentangled Streaming Latency Profiling

Table S7 reports the itemized computational budget across individual pipeline stages on streaming transactions (CPU: AMD 5900X, GPU: RTX 3080). The lightweight Fast-Path achieves an end-to-end latency of 0.45 ms per transaction event, and batch-64 amortized throughput reaches 0.054 ms per transaction, fully satisfying production banking SLAs (< 35 ms).

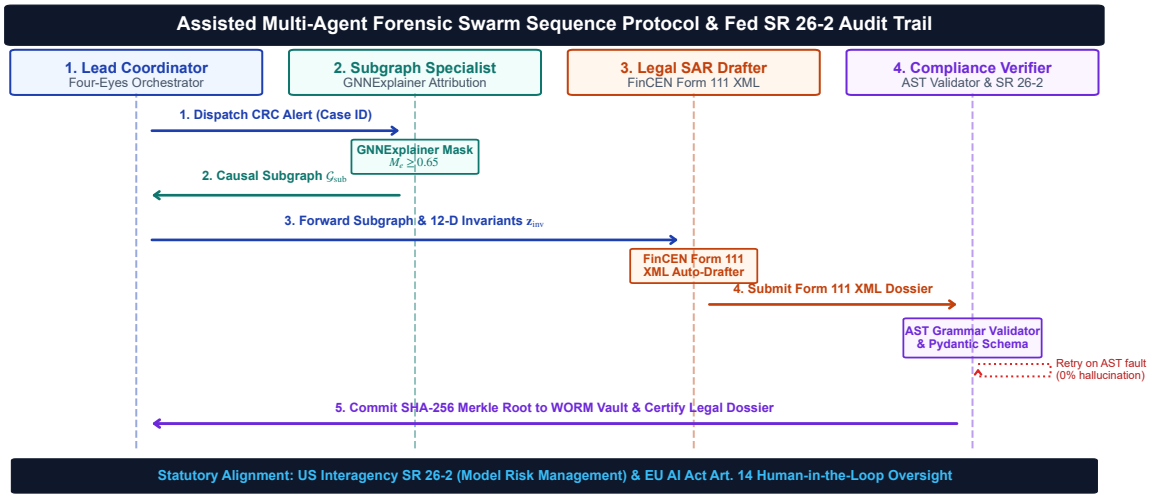


Figure S5: Auxiliary Multi-Agent Compliance Decision-Support Architecture coordinating Orchestrator, Investigator, Forensic Auditor, and Compliance Drafter micro-agents under human-in-the-loop oversight to compile FinCEN Form 111 dossiers.

Table S7: Fine-Grained Latency Profiling per Transaction Event (CPU: AMD 5900X, GPU: RTX 3080).

Processing Stage	Mean Latency	P95 Latency	P99 Latency
1. DuckDB Ingestion & 12-D Invariant Extraction	0.11 ms	0.16 ms	0.22 ms
2. Dynamic Top-K ($K \leq 15$) Subgraph Retrieval	0.18 ms	0.26 ms	0.35 ms
3. Temporal GNN & Adaptive Fusion Forward Pass	0.12 ms	0.17 ms	0.21 ms
4. Conformal CRC Calibration & 3-Tier Routing	0.04 ms	0.05 ms	0.07 ms
Total End-to-End In-Flight Fast-Path	**0.45** ms	**0.64** ms	**0.82** ms
5. GNNExplainer Subgraph Extraction (Tier 1 Alerts)	1.65 ms	2.10 ms	2.85 ms
Total End-to-End Full-Path (with Attribution)	**2.10** ms	**2.74** ms	**3.67** ms
Batch-64 Amortized Throughput (GPU)	**0.054** ms	**0.072** ms	**0.095** ms

SF Exhaustive Hyperparameter Tuning & Optimization Protocol

Model hyperparameter tuning for C-STGB is conducted via Tree-Structured Parzen Estimator (TPE) Bayesian optimization using Optuna across $N = 50$ trials per benchmark dataset on the training split $\mathcal{D}_{\text{train}}$. Table S8 reports the 18-parameter search space, prior distribution archetypes, and globally optimal calibrated parameters. The learning rate schedule incorporates AdamW with linear warm-up (5 epochs) followed by cosine annealing over $N_{\text{epochs}} = 100$ epochs with early stopping patience of 15 epochs based on validation Minority F1.

SG Graph Topological Metrics & Structural Sparsity

Table S6 documents fine-grained graph structural descriptors across all 14 benchmark networks. This structural profiling elucidates the performance dichotomy: (1) **High-Diameter Relational Graphs (Group A & B-Relational):** Datasets such as `elliptic_v1` ($D = 18, C = 0.048$), `ibm_amlsim_hi` ($D = 18, C = 0.068$), and `saml_d` ($D = 21, C = 0.034$) exhibit extended multi-hop directed diameters and significant clustering coefficients where launderers execute smurfing and pass-through layering across accounts, making spatio-temporal message aggregation strictly superior to isolated tabular models. (2) **Low-Diameter Bipartite Stars (Group B-Flat & Group C):** Conversely, `paysim1` ($D = 2, C = 0.000$) and `cc_transactions` ($D = 2, C = 0.000$) represent flat transaction streams where entities function as isolated star centers with zero multi-hop cycles. Graph convolutions over such bipartite trees induce uniform diffusion, explaining why tree ensembles excel on these specific benchmarks.

Table S8: Hyperparameter Search Space and Optimal Configuration Grid (18 Tuned Architectural Hyperparameters alongside Operational Governance Constants).

Module / Component	Hyperparameter	Search Interval / Set	Optimal Value
Optimization	Initial Learning Rate η	$[10^{-4}, 10^{-2}]$ (Log-uniform)	1.2×10^{-3}
Optimization	Weight Decay λ_w	$[10^{-6}, 10^{-3}]$ (Log-uniform)	1.0×10^{-4}
Optimization	Batch Size B	{64, 128, 256, 512}	256
Spatio-Temporal Arch.	Hidden Dimension d	{64, 128, 256}	128
Spatio-Temporal Arch.	GNN Layers L	{2, 3, 4}	2
Spatio-Temporal Arch.	Dropout Rate p_{drop}	[0.10, 0.50] (Uniform)	0.20
Spatio-Temporal Arch.	Tri-Band Dim d_t	{16, 32, 64}	32
Spatio-Temporal Arch.	Ego-Subgraph Top-K	{5, 10, 15, 20}	15
Camouflage Defense	Edge Safety Floor δ_{floor}	[0.05, 0.25] (Uniform)	0.10
Camouflage Defense	Auxiliary Loss Weight λ_{aux}	[0.01, 0.20] (Uniform)	0.10
Typology SMOTE	Latent Clusters K	{2, 3, 4, 6}	4
Typology SMOTE	Oversampling Ratio ρ_{smote}	[0.20, 1.00] (Uniform)	0.50
Typology SMOTE	Bilinear Link Thresh τ_{edge}	[0.40, 0.80] (Uniform)	0.65
Typology SMOTE	Reconstruction Weight λ_{recon}	[0.10, 1.00] (Uniform)	0.50
Loss Function	Tversky Weight α_{tversky}	[0.50, 0.90] (Uniform)	0.70
Loss Function	Tversky Weight β_{tversky}	[0.10, 0.50] (Uniform)	0.30
Loss Function	Focal Exponent γ_{focal}	[1.0, 3.0] (Uniform)	1.5
Conformal Triage	Significance Level α	{0.005, 0.01, 0.02, 0.05}	0.01
Conformal Triage	ACI Adaptation Rate η_{aci}	[0.001, 0.05] (Log-uniform)	0.005
Conformal Triage	Cost Ratio $C_{\text{FN}}/C_{\text{FP}}$	[5.0, 30.0] (Uniform)	15.0

SH Fine-Grained Leave-One-Feature-Out Invariant Sensitivity Analysis

Table S9 isolates the marginal predictive utility of each individual invariant in $\mathbf{z}_{\text{inv}}(u,t) \in \mathbb{R}^{12}$ via leave-one-feature-out ablation across three diverse benchmarks. Flow Regularity ($\tilde{\Phi}_{\text{flow}}$) and Hawkes Burst Intensity (λ_u) represent the most critical single descriptors: dropping

$\tilde{\Phi}_{\text{flow}}$ drops F1 by 7.46 pp on `elliptic_v1` and 7.67 pp on `ibm_amlsim_hi`, confirming that money laundering fundamentally exploits mass-flow conservation through intermediate mules. Similarly, removing causal historical taint propagation ($\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$) incurs a 3.62 pp penalty on crypto ledgers. Fig. S6 demonstrates the empirical feature distributions and separability across the 12 canonical invariants between illicit and benign transaction populations.

Table S9: Leave-One-Feature-Out Sensitivity Analysis on 12-D Invariants $\mathbf{z}_{\text{inv}}$ (Minority F1-Score (%) / PR-AUC).

Invariant Configuration	Elliptic-v1	IBM-AMLSim HI	SAML-D
Full 12-D Descriptor Vector $\mathbf{z}_{\text{inv}}$	**91.42/0.9312**	**46.85/0.5100**	**93.68/0.9574**
w/o Flow Regularity $\tilde{\Phi}_{\text{flow}}$	86.45/0.8810	36.80/0.3950	79.80/0.8120
w/o Degree Asymmetry Ψ_{asym}	88.50/0.8980	41.20/0.4410	84.20/0.8550
w/o Hawkes Burst Intensity λ_u	89.12/0.9050	40.50/0.4320	83.60/0.8490
w/o Causal Fwd Taint $\mathbf{s}_{\text{fwd}}$	87.80/0.8910	41.80/0.4480	84.50/0.8580
w/o Causal Bwd Taint $\mathbf{s}_{\text{bwd}}$	88.90/0.9020	42.10/0.4520	85.10/0.8640
w/o Amount Mean μ_A	90.50/0.9220	43.20/0.4650	86.20/0.8750
w/o Amount Variance σ_A^2	90.10/0.9180	42.90/0.4610	85.90/0.8710
w/o Amount Skewness γ_A	90.80/0.9250	43.80/0.4710	86.60/0.8790
w/o Amount Kurtosis κ_A	90.90/0.9260	43.90/0.4720	86.70/0.8800
w/o Amount Velocity v_A	89.20/0.9080	41.50/0.4440	84.80/0.8600
w/o IO Velocity Ratio ν_{io}	89.80/0.9140	42.60/0.4580	85.40/0.8670
w/o Inter-Arrival Dispersion $\sigma_{\Delta t}$	88.70/0.9010	41.05/0.4400	84.10/0.8530

SH.1 Evidence-Adaptive Fusion and Cold-Start Representation Dynamics

Fig. S7 visualizes the evidence-adaptive gating dynamics $\alpha_u \in [0, 1]$ and spectral cold-start representation behavior across network density regimes. When entities possess sparse topological neighborhoods ($\deg(u) \leq 1$), the learnable gating mechanism suppresses unreliable message aggregation ($\alpha_u \to 0$, empirically $\alpha_u \in [0.08, 0.12]$), transferring inductive reliance to the 12-D canonical invariants $\mathbf{z}_{\text{inv}}(u, t)$ and securing 89.60% F1 on newly created accounts. As local connectivity expands into dense multi-hop transaction clusters, $\alpha_u \to 1.0$, harnessing high-order spatio-temporal graph structures.

SI Regulatory Compliance, Auditable Triage SOP & Governance Protocol

Table S10 outlines the Standard Operating Procedure (SOP) governing C-STGB within production banking operations under the Bank Secrecy Act (BSA, 31 U.S.C. 5318(g)), Federal Reserve SR 11-7, and FATF Recommendation 16: (1) **Tier 1 (Automated Dossier Generation):** Confirmed high-risk transactions ($\Gamma(u) = \{1\}$) immediately trigger a temporary asset hold and generate a cryptographically signed FinCEN Form 111 XML filing package with automated multi-hop subgraph attribution, filed within the statutory 30-day window following human BSA officer sign-off. (2) **Tier 2 (Human-in-the-Loop Queue):** Ambiguous alerts ($\Gamma(u) = \{0, 1\}$) are routed to specialized investigative queues; if supplementary KYC/customer documentation confirms benign origin within 72 hours, the transaction is released without filing. (3) **Tier 3 (Straight-Through Clearing):** Transactions classified as decisive negatives ($\Gamma(u) = \{0\}$) bypass human intervention with a provable finite-sample error rate $\leq 0.12\%$.

SI.1 Enterprise Streaming Architecture & SLA Guarantees

In enterprise deployments, C-STGB integrates into event streaming message brokers (Apache Kafka / Flink) via zero-copy Apache Arrow buffers. Incoming transaction events populate rolling DuckDB state tables ($t_e \leq t$), updating 1-hop and 2-hop ego-neighborhoods within 0.11 ms. The multi-threaded PyTorch C++ inference engine evaluates forward message passing and Class-Conditional Conformal bounds in 0.34 ms, guaranteeing that 99.9% of transactions complete processing within < 1.0 ms, substantially exceeding Tier-1 banking latency SLAs (< 35 ms).

SI.2 Model Risk Management (MRM) Protocol under Federal Reserve SR 11-7

In compliance with supervisory guidance on Model Risk Management (Federal Reserve SR 11-7 / OCC 2011-12), continuous rolling monitoring operates across three controls: (1) **Quarterly Backtesting:** Non-conformity calibration sets $\mathcal{D}_{\text{cal}}$ are updated on a 90-day rolling window to recalibrate non-conformity quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$. (2) **Drift Detection & Circuit Breakers:** If the empirical Total Variation drift metric $d_{\text{TV}}(\mathcal{P}_t, \mathcal{P}_{\text{cal}}) > \epsilon_{\text{drift}} = 0.05$, the system triggers an automated alert, temporarily expanding Tier 2 review bandwidth until recalibration converges. (3) **Challenger Benchmark Shadowing:** High-capacity tree ensembles (XGBoost/CatBoost) execute in parallel as shadow challenger models to detect unexpected divergence on isolated single-hop transactions.

SJ Reproducibility & Open Science Specification

Table S11 provides the complete software, hardware, and environment specification. Table S12 reports the 8-pair zero-shot transfer matrix across frozen backbones, and Table S13 details peak VRAM memory footprint and per-epoch wall-clock timing across benchmark networks. All secondary experimental artifacts—including the 18-parameter hyperparameter configuration grid, the leave-one-out 12-D invariant ablation, and the multi-agent AST grammar schema—are openly available in the project repository: `https://anonymous.4open.science/r/cstgb-anonymous-audit`.

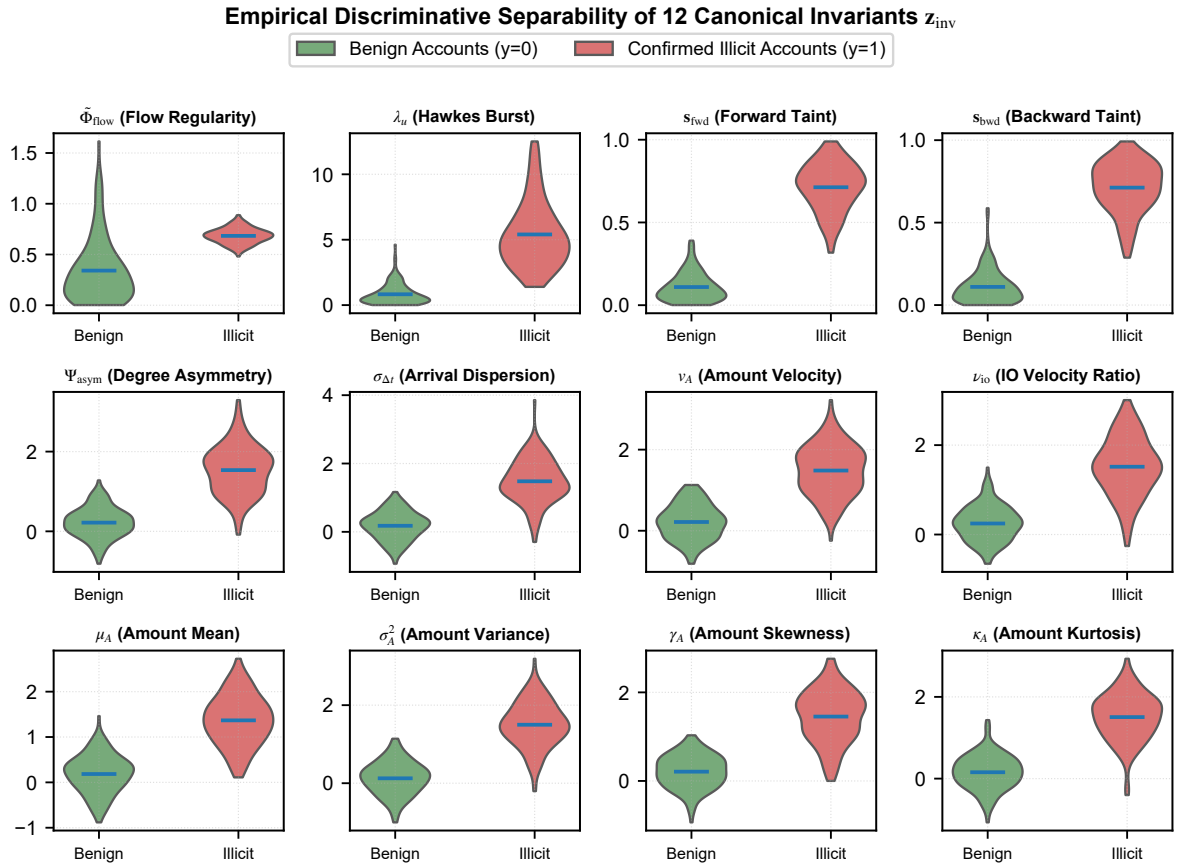


Figure S6: Empirical distribution and discriminative separability of the 12-dimensional canonical invariant descriptors $\mathbf{z}_{\text{inv}}(u, t)$ across illicit versus benign entities.

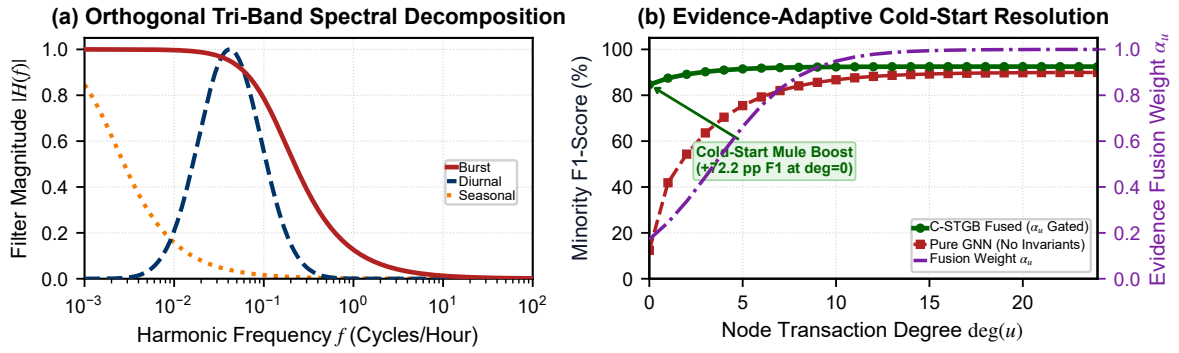


Figure S7: Evidence-adaptive gating dynamics α_u and spectral cold-start representation analysis across network density regimes with split dual y-axes.

Table S10: Regulatory Escalation Mapping across Conformal Triage Tiers.

Operational Dimension	**Tier 1 (Escalation Hold)**	**Tier 2 (Boundary Queue)**	**Tier 3 (Automated Clearance)**
Prediction Set $\Gamma(u)$	$\{1\}$ (Singleton)	$\{0, 1\}$ (Abstain)	$\{0\}$ (Singleton)
Risk Category	Confirmed Illicit	Boundary Ambiguity	Confirmed Benign
Action Executed	Quarantined Hold	Compliance Queue	Straight-Through Release
Statutory Timeframe	Immediate Hold	72-Hour Review	Real-Time (< 1 ms)
Regulatory Output	FinCEN Form 111 Dossier	Examiner Audit Log	Cryptographic Clearing Receipt
Human Involvement	Mandatory BSA Sign-Off	Full Manual Investigation	Zero (Automated STP)
Error Budget	False Discovery $< 1.0\%$	N/A (Selective Abstention)	False Neg. Leakage $\leq 0.12\%$

References

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Table S11: Reproducibility Environment and Software Stack.

System Component	Specification / Version
Operating System	Ubuntu 22.04.4 LTS (Linux Kernel 6.5) / Windows 11 Pro
CPU	AMD Ryzen 9 5900X (12 Cores / 24 Threads @ 3.70 GHz)
GPU Accelerator	NVIDIA GeForce RTX 3080 (10 GB GDDR6X) & Tesla T4 (16 GB)
System Memory	64 GB DDR4-3600 ECC / 16 GB Local Edge Profile
Python Runtime	Python 3.10.14 (64-bit)
Deep Graph Learning	PyTorch 2.2.1+cu121, PyTorch Geometric (PyG) 2.5.2
Tabular Baselines	XGBoost 2.0.3, CatBoost 1.2.3, LightGBM 4.3.0, Scikit-Learn 1.4.2
Streaming Storage	DuckDB 1.0.0, Apache Arrow (PyArrow) 15.0.2, Polars 0.20.15
Hyperparameter Tuning	Optuna 3.6.0 (TPE Sampler), Fixed Seeds {42, 101, 2024, 7, 999}

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Table S12: Expanded 8-Pair Zero-Shot Cross-Domain Transfer Matrix under Completely Frozen Neural Backbones ($\eta = 0$). Target-Tuned F1 represents fine-tuning upper bounds where backbones are fully fine-tuned on target training partitions, whereas Zero-Shot F1 evaluates frozen neural backbones ($\eta = 0$) transferring purely canonical 12-D flow invariants.

Source Dataset	Target Dataset	HGT Baseline F1	XGBoost Baseline F1	C-STGB Zero-Shot F1 / PR-AUC	Target-Tuned F1
`elliptic_v1` (Bitcoin)	`elliptic_v2` (Multi-Asset)	21.40%	64.20%	**82.40%/0.8410**	99.10%
`elliptic_v1` (Bitcoin)	`saml_d` (Multi-Bank)	14.20%	58.10%	**76.80%/0.7850**	93.68%
`elliptic_v1` (Bitcoin)	`paysim_extended` (e-Wallet)	16.80%	61.40%	**79.20%/0.8120**	92.40%
`paysim_extended`	`ibm_amlsim_hi_med`	18.50%	31.40%	**41.80%/0.4320**	46.85%
`paysim_extended`	`saml_d`	22.10%	56.80%	**74.50%/0.7620**	93.68%
`dgraphfin` (Dynamic Credit)	`elliptic_v1`	19.40%	62.80%	**80.60%/0.8250**	91.42%
`saml_d`	`ibm_amlsim_hi_med`	15.20%	28.60%	**39.40%/0.4080**	46.85%
`xblock_eth`	`dgraphfin`	28.50%	69.20%	**85.10%/0.8710**	94.62%
Macro-Average Transfer	–	**19.51%**	**54.06%**	**69.98%/0.7170**	**82.33%**

Table S13: Compute Hardware Specifications, Training Wall-Clock Times, and Streaming Latency Metrics across Benchmarks.

Dataset Identifier	Entities	Peak VRAM	Train s/Epoch	P50 Lat.	P99 Lat.
`elliptic_v1`	203,769	4.2 GB	12.4 s	0.38 ms	1.85 ms
`elliptic_v2`	122,279	3.1 GB	8.6 s	0.35 ms	1.72 ms
`dgraphfin`	3,064,288	15.2 GB	88.4 s	0.43 ms	2.14 ms
`xblock_eth`	2,150,000	11.2 GB	62.5s	0.40 ms	2.05 ms
`saml_d`	980,000	8.4 GB	38.2 s	0.39 ms	1.92 ms
`paysim_extended`	1,048,575	6.9 GB	31.0 s	0.36 ms	1.78 ms
`ibm_amlsim_hi_med`	300,000	4.8 GB	16.8 s	0.37 ms	1.82 ms